DATA ITEM DESCRIPTION

Title: HEMP SYSTEM VERIFICATION TEST PLAN

Number: DI-EMCS-81846A
AMSC Number: 10407
DTIC Applicable: No
Preparing Activity: DS

Approval Date: 20230320
Limitation: N/A
GIDEP Applicable: No
Project Number: EMCS-2023-002

Use/Relationship:

The HEMP System Verification Test Plan applies to MIL-STD-3023A, *High-Altitude Electromagnetic Pulse (HEMP) Protection for Military Aircraft*. DI-EMCS-81846, *HEMP Test Plan for System Verification Test*, is hereby cancelled and superseded by DI-EMCS-81846A. This data item description (DID) contains the format, content, and intended-use information for the data deliverable resulting from the verification test tasks described in MIL-STD-3023A sections 4.7.3 and 5.7.3 and in Appendix D, as applicable.

The purpose of verification-testing the aircraft's HEMP protection subsystem (HPS) is to demonstrate that the installed protection provides the required HEMP hardness. These tests are performed after the HPS has passed performance testing (in accordance with Appendix C of MIL-STD-3023A) and after the equipment is installed and functioning to determine if the operational system suffers mission-aborting damage or upset due to simulated HEMP excitations. Verification is normally a Government-conducted test and is not part of an aircraft construction contract.

The HEMP System Verification Test Plan describes the test-execution, data-recording, analysis, inspection, and reporting procedures to be used by the contractor to perform the verification testing. The plan provides the means for the Government to understand and duplicate methods used by the contractor to demonstrate the HPS' compliance with respect to MIL-STD-3023A's verification test pass and fail requirements. The Department of Defense's (DoD's) HEMP certification authority requires that a documented verification test plan be submitted as part of a HEMP certification request package for an aircraft.

If a portion of the HPS' components has already been tested by a previous test program, then the documentation of those previous tests, including plans, reports, and test data, is attached as an appendix to the HEMP System Verification Test Plan.

This DID supersedes DI-EMCS-81846.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and any applicable amendments, notices, and revisions, shall be as specified in the contract. The HEMP System Verification Test Plan's classification shall be in accordance with any relevant aircraft- or system-specific classification guides and also the Defense Threat Reduction Agency (DTRA) Security Classification Guide for DoD Electromagnetic Pulse

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(EMP) Programs and Activities. (DTRA's security classification guide is available by mail request to ATTN: RD-NTSA, Defense Threat Reduction Agency, 8725 John J. Kingman Road, MSC 6201, Fort Belvoir, VA 22060-6201.)

2. Format. Contractor format is acceptable.

3. Content. The HEMP System Verification Test Plan shall document the test execution, data recording, analysis, inspection, and reporting procedures to be used to comply with section 5.7.3 and Appendix D of MIL-STD-3023A, including:

a. Active system testing (AST)

b. Passive system testing (PST)

c. Internal direct drive test (DDT)

d. External direct drive test (DDT)

3.1 Administrative data. The HEMP System Verification Test Plan shall contain an administrative section covering the following:

a. Contract number, contract data requirements list (CDRL) number, and date

b. Authentication and certification by a qualified representative of the procuring activity for the test plan

3.2 Summary information. The HEMP System Verification Test Plan shall summarize the following:

3.2.1 Introduction, including:

a. General objectives of the verification test

b. A general description of the HPS' topology to be tested

c. A statement of any assumptions or limitations associated with the planned verification testing

d. A summary of the verification test's pass and fail requirements per MIL-STD-3023A

3.2.2 Scope. A general description of the overall test-requirements matrix that will be used to verify the HPS' compliance with MIL-STD-3023A's performance requirements, including the relative roles of tests, inspections, and analyses.

3.2.3 Methods of verification. A general description of the procedures that will be used to verify the HPS' compliance with each verification test performance requirement as listed in 3.3 below.

3.2.4 Engineering factors. Any important engineering factors that may affect the verification test procedures, such as: physical accessibility constraints of the aircraft system's topology, resource limitations, safety and security protocols, and precedent reports.

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3.3 Detailed information. The overall HEMP System Verification Test Plan shall describe in detail the test procedures and the associated test equipment. The HEMP System Verification Test Plan shall also include four separate subsections describing, respectively, the detailed plans for the AST, PST, internal DDT, and external DDT execution.

3.3.1 The overall test plan. The main portion of the test plan shall contain the following detailed information:

a. An identification of the aircraft to be tested, including tail number (if determined), photographs, figures, and diagrams

b. A physical and functional description of the aircraft

c. Identification of the proposed test team organizations and personnel, and of the Government organizations and witnesses that will likely be present

d. A description of the HPS' topology, including a description of each HEMP protection measure and drawings showing the locations of the following items:
   i. all points of entry (POEs) and penetration protective devices (PPDs) of each electromagnetic (EM) barrier
   ii. equipment layouts inside each EM barrier
   iii. the installation of mission critical systems (MCS) outside of any EM barrier and their associated special protective measures (SPMs)
   iv. cable interconnections external to each EM barrier

e. Descriptions of the data acquisition equipment intended to be used, including: the manufacturer, model, and serial numbers of each equipment item, detailed calibration procedures for the equipment, calibration-traceability documentation, and a layout showing the interconnectivity of the overall data acquisition instrumentation system

f. Descriptions of the test procedures to be used, including system, subsystem, and circuit configuration requirements; equipment operating states, test configuration diagrams; test point locations and descriptions; and functional monitoring provisions

g. Descriptions of methods for measuring noise (instrument inherent and local or ambient)

h. Descriptions of the models, techniques, and tools used for preliminary analyses and predictions, and their specific application to the specific aircraft to be tested

i. Descriptions of the methods that will be applied to quantify and evaluate any failures of the HPS' performance with respect to all pass and fail criteria

j. Identification of the pass and fail criteria, and of any deviations from the MIL-STD-3023A requirements that are anticipated, along with an explanation of why these deviations may be necessary

k. Descriptions of the data-management procedures that will be used, including: data quality control procedures, data acceptability criteria, annotation and preservation of

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data records, and the methods to compare results to pass and fail criteria

l. Identification of any potential safety risks, and descriptions of procedures to mitigate these risks, including any EM radiation hazards and electrical shock hazards

m. Descriptions of the data security and physical security procedures that will be applied

n. A description of the planned test schedule, including a listing of the prioritization of measurements

o. A description of the methods that will be used to simulate the operational performance of any MCS during the test if actual operation of that MCS during the test would be impractical

p. A written confirmation that the required MIL-STD-461 immunity testing of mission critical systems (MCS) was, or will be, completed, with results available prior to the verification test, and a table summarizing available immunity data for each MCS

q. Descriptions of how the raw and processed test data will be recorded, and the electronic data format that will be used to store the data

r. A discussion of any anticipated uncertainties involved with the planned test procedure

3.3.2 The AST portion of the test plan. This portion of the HEMP System Verification Test Plan shall contain the following detailed information:

a. Detailed test procedures in accordance with Appendix D, section D.5, of MIL-STD-3023A

b. Descriptions of the AST antenna system, including key performance parameters

c. Description of the antenna's field map characteristics, the methods that will be used to monitor the incident field and its relationship to these field map characteristics, and discussion of any anticipated uncertainties involved in the HEMP incident field measurements

d. Identification, with diagrams, of the planned placement of the aircraft under test in the AST simulator's working volume and the planned placement of other system elements that will be required for achieving functionality, with respect to the simulator working volume

e. Description of the planned chronology of the AST execution, including the planned sequence of events with aircraft orientation relative to the HEMP electric field and corresponding polarization of the HEMP electric field

f. Identification of the aircraft's functional configurations that will be tested

g. Identification of the types of functional checks that will be made before, during, and after pulsing

h. Identification of the following planned instrumentation locations, with an explanation for why these specific points are proposed:

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i. Reference sensor

ii. Other measurement points as required

3.3.3 The PST portion of the test plan. This portion of the HEMP System Verification Test Plan shall contain the following detailed information:

a. Detailed test procedures in accordance with Appendix D, section D.5, of MIL-STD-3023A

b. Descriptions of the PST HEMP threat-level simulator to be used (including key performance parameters)

c. Description of the antenna's field map characteristics, the methods to be used to monitor the incident field and the incident field's relationship to these field map characteristics, and discussion of any anticipated uncertainties involved in the HEMP incident field measurements

d. Identification, with diagrams, of the planned placement of the aircraft under test in the PST HEMP threat-level simulator's working volume, and of the planned placement of other system elements that will be required for achieving functionality (such as ground-power sources), with respect to the simulator working volume

e. Description of the planned chronology of the PST execution, including the planned sequence of events with aircraft orientation relative to the horizontally or vertically polarized HEMP threat-level simulator

f. Identification of the intended functional states of every MCS during the testing

g. Diagrams of the planned interconnecting cable layouts

h. A table listing all planned test point locations and a description of how these test points will relate to the MCS locations and to the hardness allocation process for the aircraft

i. Identification of the types of functional checks that will be made before and after pulsing

j. Identification of the following planned instrumentation locations, with an explanation for why these specific points are proposed:

i. Electric and magnetic field sensors

ii. Reference field sensor

iii. Representative MCS cable measurement test points, both within and external to all EM barriers

iv. Other measurement points as required

k. Discussion of any anticipated uncertainties involved in the PST measurements

l. Identification of the measurement-range sensitivities of the field probes, current probes, and data acquisition system to be used in the test

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m. Describe in detail how the PST test data will be used to generate the DDT waveforms

3.3.4 The internal DDT portion of the test plan. This portion of the HEMP System Verification Test Plan shall contain the following detailed information:

a. Detailed test procedures in accordance with Appendix D, section D.5, of MIL-STD-3023A

b. Identification, with diagrams, of the locations of each HPS component that will be subjected to internal DDT injection and a description of how these test points will relate to the MCS and to the hardness allocation process for the aircraft

c. Description of the planned chronology of the internal DDT execution, including the planned sequence of events with the upper-bound injection current waveforms for each test point measured in the internal DDT

d. Identification of the transfer impedance of the current-measurement probes to be used in the test, and measurement range of the end-to-end data acquisition system to be used in the test

e. Description of how the internal DDT pulser and current probes will be connected and their performance monitored

f. Discussion of any anticipated uncertainties involved in the internal DDT measurements

g. Descriptions of the methods that will be used to determine the onset of any MCS upset or damage, and the locations for which observations will be made

h. Identification of any special requirements of the test facility's operations that may impede HEMP testing

3.3.5 The external DDT portion of the test plan. This portion of the HEMP System Verification Test Plan shall contain the following detailed information:

a. Detailed test procedures in accordance with Appendix D, section D.5, of MIL-STD-3023A

b. Identification, with diagrams, of the locations of each HPS component that will be subjected to external DDT injection and a description of how these test points will relate to the MCS and to the hardness allocation process for the aircraft

c. Description of the planned chronology of the external DDT execution, including the planned sequence of events with the upper-bound injection current waveforms for each test point measured in the external DDT

d. Identification of the transfer impedance of the current-measurement probes to be used in the test, and measurement range of the end-to-end data acquisition system to be used in the test

e. Description of how the external DDT pulser and current probes will be connected

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and their performance monitored

f. Discussion of any anticipated uncertainties involved in the external DDT measurements

g. Descriptions of the methods that will be used to determine the onset of any MCS upset or damage, and the locations for which observations will be made

h. Identification of any special requirements of the test facility's operations that may impede HEMP testing

3.4 Other information sources. When other available information sources contain data required by the HEMP System Verification Test Plan, these sources shall be referenced, rather than duplicated, within the test plan.

End of DI-EMCS-81846A

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